

Kevin R Sitek

Research Assistant Professor, Northwestern University

✉ kevin.sitek@northwestern.edu 🌐 sitek.github.io 🎓 Google Scholar 🆔 0000-0002-2172-5786
 🏠 Kevin-Sitek 🔄 sitek 🦋 sitek.bsky.social 📧 @sitek@fediscience.org

Education

- PhD** **Harvard University**, Speech and Hearing Bioscience and Technology 2013 – 2019
- Division of Medical Sciences, Harvard–MIT Program
 - Advisor: Satrajit S. Ghosh, PhD
 - Dissertation: [Mapping the Human Subcortical Auditory System With MRI](#)
- BA** **University of California, Berkeley**, Linguistics (Honors) and Cognitive Science 2007 – 2011
- Advisor: Keith Johnson, PhD
 - Linguistics honors thesis: Phonetic context effects in ipsilateral and contralateral conditions

Experience

- Northwestern University**, Research Assistant Professor Evanston, IL 2023 – present
- Department of Communication Sciences and Disorders
 - Affiliated Faculty: Cognitive Science Program; Northwestern University Interdepartmental Neuroscience (NUIN) PhD Program
- University of Pittsburgh**, Research Scientist Pittsburgh, PA 2021 – 2023
- Department of Communication Science and Disorders
 - SoundBrain Lab (PI: Bharath Chandrasekaran, PhD)
- University of Pittsburgh**, Postdoctoral Associate Pittsburgh, PA 2021 – 2021
- Department of Communication Science and Disorders
 - SoundBrain Lab (PI: Bharath Chandrasekaran, PhD)
- Baylor College of Medicine**, Postdoctoral Associate Houston, TX 2019 – 2021
- Department of Neuroscience
 - High Resolution Brain Imaging Lab (PI: David Ress, PhD)
- MIT McGovern Institute for Brain Research**, Graduate Student Researcher Cambridge, MA 2014 – 2019
- Gabrieli Laboratory
- San Francisco VA Medical Center / NCIRE**, Staff Research Assistant I/II San Francisco, CA 2011 – 2013
- Brain Imaging and EEG Laboratory (PIs: Judith Ford, PhD; Daniel Mathalon, PhD, MD)
- UC Berkeley Department of Linguistics**, Undergraduate/Staff Research Assistant Berkeley, CA 2009 – 2011
- Phonology Laboratory (PI: Keith Johnson, PhD)

Grants

- Auditory–motor interactions in neural processing and the frequency-following response** 2025–2028
- NIH NIDCD ECR R21 · Grant: 1R21DC022906-01
 - Role: Principal Investigator
 - Co-Investigators: Bharath Chandrasekaran, PhD; Peer Herholz, Dr. rer. nat.

High-resolution functional imaging of speech-induced sensory modulation • NIH NIDCD R01 · Grant: R01 DC020963-01A1 • Role: Co-Investigator (PI: Jason W. Bohland, PhD)	2024–2028
Investigating human non-lemniscal inferior colliculus contributions to auditory learning with 7T MRI • NIH NIDCD K01 Mentored Career Development Award · Grant: K01 DC019421-01A1 • Role: Principal Investigator • Co-Mentors: Bharath Chandrasekaran, PhD; Tamer Ibrahim, PhD; Srivatsun Sadagopan, PhD	2022–2027
Administrative Supplement — K01 DC019421 • NIH NIDCD Administrative Supplement · Grant: 3K01DC019421-05S1 • Role: Principal Investigator	2025–2026
Cortical-subcortical interactions in auditory cognition • CIHR Project Grant • Role: Collaborator (PI: Robert Zatorre, PhD)	2023–2028
Subcortical auditory feedback mechanisms in speech: Function and structure • NIH NIDCD F31 NRSA Predoctoral Fellowship · Grant: F31 DC015695-01 (completed) • Role: Principal Investigator • Sponsor: Satrajit S. Ghosh, PhD; Co-Sponsor: John D. E. Gabrieli, PhD	2016–2018
Submitted Manuscripts and Preprints	
Motor signals modulate cortical but not subcortical processing of self-initiated sounds Raiff, L., Butler, G., McFarlane, K., Chandrasekaran, B., Sitek, K.R. 10.64898/2026.07.10.737812 (bioRxiv)	in review
Evaluating Approaches for Inference Testing of Whole-Brain Densely Sampled Single-Subject Task fMRI Data Medina, M.C., Reddy, N.H., Bright, M.G., Sitek, K.R. 10.64898/2026.06.29.735344 (bioRxiv)	in review
Distinct 7T functional MRI response patterns to stimulus, response, and feedback across dorsal striatum during auditory learning Sitek, K.R. , Roark, C.L., Helou, L.B., Chandrasekaran, B.	in revision
The recognition and comprehension of speech sound patterns show a mechanistic dissociation along the ventral and dorsal streams Llanos, F., Sitek, K.R. , Feng, G., Chandrasekaran, B.	in revision
Auditory corticostriatal connections in the human brain Sitek, K.R. , Helou, L.B., Chandrasekaran, B. 10.1101/2022.08.04.502679 (bioRxiv)	in revision
Bayesian Tensor Factorized Mixed Effects Vector Autoregressive Processes for Inferring Granger Causality Patterns from High-Dimensional Neuroimage Data Fan, J., Sitek, K.R. , Chandrasekaran, B., Sarkar, A. 10.48550/arXiv.2206.10757 (arXiv)	2022

Publications

- Mapping Whole-Brain Auditory Activation with 3T Multi-Echo fMRI at the Group and Individual-Subject Level** 2026
Medina, M.C., Reddy, N.H., **Sitek, K.R.**, Bright, M.G.
[10.1016/j.heares.2026.109622](https://doi.org/10.1016/j.heares.2026.109622) (Hearing Research)
- Communication is the foundation of an impactful and resilient scientific community** 2025
Sitek, K.R., Wearn, A., DuPre, E.M.
[10.52294/001c.143510](https://doi.org/10.52294/001c.143510) (Aperture Neuro 2025;5(SI 3))
- The Organization for Human Brain Mapping Time Machine: A freely accessible archive of Annual Meeting talks on YouTube** 2025
Wearn, A., **Sitek, K.R.**, Valk, S.L., Forkel, S.J.
[10.52294/001c.138647](https://doi.org/10.52294/001c.138647) (Aperture Neuro 2025;5(SI 2))
- Functional connectivity across the human subcortical auditory system using an autoregressive matrix-Gaussian copula graphical model approach with partial correlations** 2024
Chandra, N.K., *Sitek, K.R.* **, Sarkar, A., Chandrasekaran, B.
[10.1162/imag_a_00258](https://doi.org/10.1162/imag_a_00258) (Imaging Neuroscience)
- Cortical-subcortical interactions to violations of auditory predictions measured with 7T functional MRI** 2024
Ara, A., Provias, V., **Sitek, K.R.**, Coffey, E., Zatorre, R.
[10.1093/cercor/bhae316](https://doi.org/10.1093/cercor/bhae316) (Cerebral Cortex)
- On the role of neural oscillations across timescales in speech and music processing** 2022
Gnanateja, G.N., Devaraju, D.S., Heyne, M., Quique, Y.M., **Sitek, K.R.**, Tardif, M.C., Tessmer, R., Dial, H.R.
[10.3389/fncom.2022.872093](https://doi.org/10.3389/fncom.2022.872093) (Frontiers in Computational Neuroscience)
- Structural connectivity of human inferior colliculus subdivisions using in vivo and post mortem diffusion MRI tractography** 2022
Sitek, K.R., Calabrese, E., Johnson, G.A., Ghosh, S.S., Chandrasekaran, B.
[10.3389/fnins.2022.751595](https://doi.org/10.3389/fnins.2022.751595) (Frontiers in Neuroscience)
- Depth relationships and measures of tissue thickness in dorsal midbrain** 2020
Truong, P., Kim, J.H., Savjani, R., **Sitek, K.R.**, Hagberg, G., Scheffler, K., Ress, D.
[10.1002/hbm.25185](https://doi.org/10.1002/hbm.25185) (Human Brain Mapping)
- Mapping the human subcortical auditory system using histology, postmortem MRI and in vivo MRI at 7T** 2019
Sitek, K.R., *Gulban, O.F.*, Calabrese, E., Johnson, G.A., Lage-Castellanos, A., Moerel, M., Ghosh, S.S.†, De Martino, F.†
[10.7554/eLife.48932](https://doi.org/10.7554/eLife.48932) (eLife)
- Alterations in the inferior longitudinal fasciculus in autism and associations with visual processing: a diffusion-weighted MRI study** 2018
Boets, B., Van Eylen, L., **Sitek, K.**, Moors, P., Noens, I., Steyaert, J., Sunaert, S., Wagemans, J.
[10.1186/s13229-018-0188-6](https://doi.org/10.1186/s13229-018-0188-6) (Molecular Autism)

Diffusion tensor imaging of central auditory pathways in patients with sensorineural hearing loss: a systematic review	2018
Tarabichi, O., Kozin, E.D., Kanumuri, V.V., Barber, S., Ghosh, S., Sitek, K.R. , Reinshagen, K., Herrmann, B., Remenschneider, A.K., Lee, D.J. 10.1177/0194599817739838 (Otolaryngology–Head and Neck Surgery)	
Decreased cerebellar-orbitofrontal connectivity correlates with stuttering severity: whole-brain functional and structural connectivity associations with persistent developmental stuttering	2016
Sitek, K.R. , Cai, S., Beal, D.S., Perkell, J.S., Guenther, F.H., Ghosh, S.S. 10.3389/fnhum.2016.00190 (Frontiers in Human Neuroscience)	
Auditory cortex processes variation in our own speech	2013
Sitek, K.R. , Mathalon, D.H., Roach, B.J., Houde, J.F., Niziolek, C.A., Ford, J.M. 10.1371/journal.pone.0082925 (PLoS ONE)	

Teaching

Communicating Science (CSD 550-3)	2024, 2025
• Instructor of Record (Adjunct), Northwestern University • Graduate PhD course: oral, written, visual, and multimedia science communication	
Speech Communication (MIT 6.541/HST.710/SHBT 204)	2015, 2016
• Graduate Teaching Assistant, MIT/Harvard University	
Guest Lectures	2016—present
• Auditory Cognitive Neuroscience — Northwestern CSD 366 (Cognitive Science Proseminar) • Subcortical Auditory Anatomy — Pitt CSD 2110 (Neuroscience of Communication) • Subcortical Contributions to Language — UT Dallas (Brain & Language) • Subcortical Auditory Anatomy and Human Function — Pitt CSD 1024 (Anatomy & Physiology of Hearing) • Acoustic Phonetics: Vowels — Harvard SHBT 204 (Speech Communication)	

Mentorship

PhD Students	
• Laura Raiff, Biomedical Engineering, Northwestern (2026–present; lead advisor, dissertation advisory committee) • Michelle Medina, Biomedical Engineering, Northwestern (2024–present; dissertation advisory committee) • Serena Mon, Communication Sciences and Disorders, Northwestern (2023–2024; qualifying exam committee)	
Doctor of Audiology	
• Laura Cahalan, University of Pittsburgh (Capstone Research Project)	

Undergraduate Research Assistants

- Max Chen, BA Cognitive Science, Northwestern (2025–present; FFRPREP contributor)
- Gabrielle Butler, BA Cognitive Science, Northwestern (2024–2025; honors thesis)
- Amp Kangusmrith, BA Cognitive Science, Northwestern (2024–present; Independent Study in Cognitive Science)
- Angelina Dinardo, BA CSD, University of Pittsburgh (Completed MS in Speech Language Pathology at Northeastern University)
- Madison Andreano, BA CSD, University of Pittsburgh (Completed her MS in Speech Language Pathology at the University of Minnesota)
- Olivia Flemm, BA CSD, University of Pittsburgh (Completed her AuD at the University of Pittsburgh)

High School Research Assistants

- Leo Chen (2025, Northwestern University; FFRPREP contributor)
- Karen Linares Mendoza (2022, University of Pittsburgh; Now pursuing undergraduate studies at Johns Hopkins University)

Awards and Honors

Aperture Neuro Best Reviewer Award	2025
• Organization for Human Brain Mapping, Brisbane, Australia	
ASHA Lessons for Success	2023
• Accepted participant	
OHBM Merit Award	2020
Harvard Brain Science Initiative Young Scientist Travel Award	2018
Mass. Eye and Ear Amelia Peabody Scholarship	2017–2018
OHBM Hackathon Travel Award	2017
• Vancouver, BC	
Neurohackweek / Neurohackademy	2017
• Accepted participant, Seattle, WA	
Certificate of Distinction in Teaching	2015
• Harvard University, Derek Bok Center for Teaching and Learning	

Service

Aperture Neuro Editorial Review Board	2026—present
Frontiers in Language Sciences Review Editor	2022—present
Ad hoc reviewer	
• Journals: <i>Aperture Neuro</i> , <i>Brain & Language</i> , <i>Brain Structure and Function</i> , <i>Cerebral Cortex</i> , <i>Frontiers in Psychology</i> , <i>Heliyon</i> , <i>Journal of the Association for Research in Otolaryngology</i> , <i>Journal of Neuroscience Research</i> , <i>Journal of Speech, Language, and Hearing Research</i> , <i>Molecular Genetics & Genomic Medicine</i> , <i>NeuroImage</i> , <i>Neuropsychologia</i> , <i>OHBM Abstract Submissions</i> , <i>PLOS ONE</i>	
• Granting agencies: European Research Council, Swiss National Science Foundation	
PhD Admissions Committee, Northwestern University Interdepartmental Neuroscience (NUIN) Program	2025–2026

ASHA Convention Topic Committee: Hearing, Balance, Tinnitus	2024, 2025
Undergraduate Studies Committee, Northwestern CSD	2023–2025
Organization for Human Brain Mapping, Communications Committee • Past Chair (2024–2025) • Chair (2023–2024) • Chair-Elect / Blog Team Lead (2022–2023) • Podcast Team Co-founder and Editor (2021–2022)	2021–2025
Organization for Human Brain Mapping, Student–Postdoc Special Interest Group • Social Media and Communications Lead	2019–2021
Society for Cognitive Auditory Neuroscience (SCAN) Moderator	2021–2023
Science in the News (Harvard) • Social Media Outreach Lead (2015–2019) • Editor, Science in the News Blog (2013–2018)	2013–2019
BrainHack Boston Organizing Committee	2017
SMALL Journal Club Co-founder and Organizer	2016–2019